A Constructive Introduction to Quantum Field Theory

Lecture 4: Φ^4_3 on the torus I. The Wetterich FBSDE, two renormalisations and the large-field problem

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Abstract

The last three lectures construct the massive Φ^4_3 measure on the three-dimensional torus with the forward–backward equations of Lecture 3. In this lecture we set up the problem. For the mass regulator the Polchinski process of Lecture 3 is the linear estimator $C_q P_q$ built from noisy observations of the field, and its forecast is their posterior mean; this gives an exact FBSDE at every lattice cutoff. Its backward half is governed by the Wetterich equation. We cannot solve that equation, so we construct an explicit approximate effective average action and write the exact FBSDE in the coordinates it defines: the *approximate forecast* Y and the *error* Z. The error is driven only by the failure of the approximate action to solve the Wetterich equation. We explain why the approximate action needs two renormalisations, in two senses: a linearly and a logarithmically divergent mass counterterm, and two correction steps beyond one loop, which make the defect integrable in the backward equation. Finally we identify the large-field problem: the construction only controls fields of size $r_q = (q/\gamma)^{1/2}$, and must be extended to all fields without losing coercivity. The price is an error that is not small but lives on large fields. Lecture 5 controls it by a priori estimates on the FBSDE, and Lecture 6 removes the cutoffs.

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Introduction

Lecture 3 formulated a programme (Figure 5.1 and the remark following it): solve the FBSDE

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t = \mathbb{E}_t[-D V_T(\Phi_T)],$$

with a renormalised interaction V_T, with estimates uniform in the ultraviolet cutoff, and take the limit. For Φ_3^4 the force F_t is not expected to converge to an integrable terminal value, and the naive estimates of the force diverge (Section 9.2). In these three lectures we carry out a version of the programme on the unit torus $\mathbb{T}^3$, for a large mass. The guiding principle is the one of Section 6: do not estimate Φ and F directly, but pass to the forecast coordinate, in which the forward equation has no drift and the tree diagrams have been integrated exactly.

The argument has three parts, one per lecture.

i. *Deterministic renormalisation* (this lecture). Construct an explicit approximate effective average action $\bar{\Gamma}_q$ and a counterterm, by a finite number of steps of the Wetterich flow, and estimate its defect.

ii. *A priori estimates* (Lecture 5). Write the exact FBSDE in the coordinates of $\bar{\Gamma}_q$, and estimate the forward and the backward equation together, uniformly in the lattice.

iii. *Removing the cutoffs* (Lecture 6). Remove the upper scale at fixed lattice, then the lattice, and identify the limiting FBSDE and its terminal law.

The emphasis is on the structure of the argument. We give complete proofs of the identities and of the scalar models of the estimates, and only describe the tensor estimates on the lattice. The lectures follow the course notes [FBSDE26], where all the estimates are proved. The point of view is that of [BCG26]; for other constructions of Φ_3^4 see [GJ73, FO76, BFS83, BG20, GH21].

What is proved. The construction is for a large mass $m^2 = q_0$ compared to the coupling, $\lambda \ll q_0^{1/2}$: a weak dimensionless coupling on a fixed torus. It gives a subsequential limit of the lattice measures, with a limiting FBSDE driven by a Brownian motion obtained as the limit of the lattice innovations. It does not prove uniqueness of the limit, the Osterwalder–Schrader axioms, or an infinite-volume limit.

11 The problem and the exact FBSDE

11.1 Lattice, scales and notation

Let N be an odd integer and $\Lambda_N = (N^{-1}\mathbb{Z}/\mathbb{Z})^3$ the periodic lattice of mesh N^{-1} in the unit torus. Let $H_N = \mathbb{R}^{\Lambda_N}$ with

$$\langle u, v\rangle = \int u\, v, \qquad \int f := N^{-3}\sum_{x\in\Lambda_N} f(x),$$

so that the volume is 1 and integrals approximate continuum integrals. The nearest-neighbour Laplacian is $\Delta_N u(x) = N^2\sum_{i=1}^3 [u(x+e_i/N) + u(x-e_i/N) - 2u(x)]$; we usually drop the index N. A kernel $C(x,z)$ acts by $Cu(x) = \int C(x,z)\,u(z)\mathrm{d}z$, so the identity has kernel $N^3 1_{x=z}$, and

$$\mathrm{Tr}C = \int C(x,x)\mathrm{d}x, \qquad (\mathrm{diag}C)(x) = C(x,x), \qquad \|C\|_{\mathrm{row}} = \sup_x \int |C(x,z)|\mathrm{d}z.$$

M_f is multiplication by f, and $K + \gamma y^2$ means $K + \gamma M_{y^2}$. For a functional F on H_N, DF, D^2F, D^kF are its gradient and derivatives, as in Lecture 3. $\|u\|_p$ is the L^p norm for the measure $\int$.

The resolution parameter. Lecture 3 used a general covariance profile. Here we use the *mass-regulator profile* of the exercise on regulators at the end of Section 6: with $m^2 = q_0$ and $t \geq 0$,

$$\hat{Q}_t(k) = \frac{1}{|k|^2 + q_0} - \frac{1}{|k|^2 + q_0 + t^2}, \qquad R_t = t^2.$$

It is convenient to parametrise the resolution by the squared regulator mass

$$q := q_0 + t^2 \in [q_0, \infty).$$

Set

$$K_q := q - \Delta_N, \qquad C_q := K_q^{-1}, \qquad C := C_{q_0} = (m^2 - \Delta_N)^{-1}. \tag{11.1}$$

In the notation of Lecture 3, with $T = \infty$ and using q instead of t as the resolution variable (so $\dot{Q}$ now means $\partial_q Q_q$),

$$Q_q = C - C_q, \qquad \partial_q Q_q = C_q^2, \qquad \dot{Q}^{1/2} = C_q, \qquad S_q = C_q, \qquad R_q = q - q_0, \qquad \partial_q R_q = 1. \tag{11.2}$$

So C_q is the covariance of the scales not yet resolved, and $K_q = S_q^{-1}$ is the precision of the posterior at resolution q under the free prior μ^C. The main properties of C_q are elementary.

Lemma 11.1. (Free resolvents) *For $q \geq 1$, uniformly in N: C_q has a nonnegative kernel, $C_q 1 = q^{-1}1$, $\|C_q\|_{p\to p} = q^{-1}$ for $1 \leq p \leq \infty$, and*

$$a_2(q) := (C_q^2)(x,x) = \mathrm{Tr}C_q^2 \lesssim q^{-1/2} \wedge N^3 q^{-2}, \qquad \mathrm{Tr}C_q \lesssim N \wedge N^3 q^{-1}. \tag{11.3}$$

For $y \in H_N$ and $\gamma > 0$, the killed resolvent $G_q(y) = (K_q + \gamma y^2)^{-1}$ satisfies $0 \leq G_q(y)(x,z) \leq C_q(x,z)$.

Proof. $C_q = \int_0^\infty \mathrm{e}^{-qs}\mathrm{e}^{s\Delta}\mathrm{d}s$ and $\mathrm{e}^{s\Delta}$ is the transition kernel of the lattice random walk. This gives positivity and the row sums. Killing the walk at rate γy^2 gives the comparison for G_q. The symbol of $-\Delta_N$ is comparable to $|k|^2$ on the centred frequency cube $n \in \{-\frac{N-1}{2}, \dots, \frac{N-1}{2}\}^3$ (symbol $4N^2\sum_i \sin^2(\pi n_i/N)$, with $k = 2\pi n$), and $\sum_k (q + |k|^2)^{-2} \leq C\int_{\mathbb{R}^3} (q + |k|^2)^{-2}\mathrm{d}k = Cq^{-1/2}$. Counting the N^3 modes gives the other bounds. □

The *bubble* $a_2(q)$ is the coincident-point variance of the shell at resolution q: $\partial_q \mathrm{Var}(X_q(x)) = a_2(q)$ for the free field at resolution q, and $\int_{q_0}^{\infty} a_2 = \mathrm{Tr} C_{q_0} \asymp N$ is the divergent variance of the lattice field.

11.2 The cutoff measure and the goal

Fix $\lambda > 0$ and put $\gamma := 3\lambda$ (not to be confused with the counterterm γ_T of Lecture 3). The interaction and the lattice measure are

$$U(\varphi) = \frac{\lambda}{4}\int \varphi^4, \qquad V_N(\varphi) = U(\varphi) + \frac{\mu_N}{2}\int \varphi^2, \qquad \nu_N(\mathrm{d}\varphi) = \frac{\mathrm{e}^{-V_N(\varphi)}}{Z_N}\,\mu^C(\mathrm{d}\varphi), \tag{11.4}$$

where μ^C is the centred Gaussian measure with covariance $C = C_{q_0}$ on H_N, and $\mu_N \in \mathbb{R}$ is a mass counterterm to be chosen. With this normalisation $D^2 U(y) = \gamma M_{y^2}$. In Lecture 3 we wrote the interaction as $\lambda \int H_4(\varphi; c_T) + \gamma_T \int H_2(\varphi; c_T)$; the Wick ordering is now part of μ_N, and the coupling is rescaled by 4. For every real μ_N, ν_N is a probability measure with all moments, since the quartic term dominates.

Theorem 11.2. (Main result, informal [FBSDE26]) *Let $\lambda > 0$ and let q_0 be large enough, depending on λ. There are explicit μ_N, with $\mu_N = -3\lambda \mathrm{Tr} C + O(\log N)$, such that:*

a) the laws ν_N, embedded in $H^{-s}(\mathbb{T}^3)$, are tight for every $s > \frac{1}{2}$;

b) along a subsequence, the forecast processes converge, on a common probability space, to a solution of a forward–backward system on $[q_0, \infty)$ driven by a cylindrical Brownian motion, whose terminal value has the limiting law ν;

c) ν satisfies the integration by parts formula of Theorem 19.5, in which the renormalised cube appears only tested in expectation.

The bounds that prove (a) and (b) are uniform in N: this is *UV stability*. They are bounds on the *error* Z of an approximate forecast, defined below. We now set up the exact FBSDE at fixed N.

11.3 The Polchinski process as a posterior

At a fixed lattice everything can be realised by a simple statistical experiment. Let $\Phi_\infty \sim \nu_N$, and let B be an independent Brownian motion on H_N, $\mathbb{E}\,\langle B_t, u\rangle\langle B_s, v\rangle = (t \wedge s)\langle u, v\rangle$. We observe

$$P_q = (q - q_0)\,\Phi_\infty + B_{q-q_0}, \qquad q \geq q_0, \tag{11.5}$$

with the filtration $\mathcal{F}_q$ generated by $(P_s)_{s \leq q}$ (completed), and $\mathbb{E}_q = \mathbb{E}[\,\cdot\,|\mathcal{F}_q]$. Let

$$\mathcal{K}_q(p) := \log \int \mathrm{e}^{\langle p,\varphi\rangle - \frac{q-q_0}{2}\|\varphi\|_2^2}\nu_N(\mathrm{d}\varphi), \qquad \nu_{q,p}(\mathrm{d}\varphi) := \mathrm{e}^{\langle p,\varphi\rangle - \frac{q-q_0}{2}\|\varphi\|_2^2 - \mathcal{K}_q(p)}\nu_N(\mathrm{d}\varphi). \tag{11.6}$$

Proposition 11.3. (Posterior, forecast and innovation) *At every fixed N:*

a) the conditional law of Φ_∞ given $\mathcal{F}_q$ is ν_{q,P_q}. In particular the forecast *and the* conditional covariance *are*

$$\psi_q := \mathbb{E}_q \Phi_\infty = D\mathcal{K}_q(P_q), \qquad \mathrm{Cov}_q := \mathrm{Cov}(\Phi_\infty|\mathcal{F}_q) = D^2\mathcal{K}_q(P_q) > 0; \tag{11.7}$$

b) the innovation $W_q := P_q - \int_{q_0}^{q} \psi_s \mathrm{d}s$ *is an $(\mathcal{F}_q)$-Brownian motion on H_N, and*

$$\mathrm{d}P_q = \psi_q\,\mathrm{d}q + \mathrm{d}W_q, \qquad \mathrm{d}\psi_q = \mathrm{Cov}_q\,\mathrm{d}W_q; \tag{11.8}$$

c) $\mathcal{K}$ solves $\partial_q \mathcal{K}_q + \frac{1}{2}\mathrm{Tr} D^2\mathcal{K}_q + \frac{1}{2}\|D\mathcal{K}_q\|_2^2 = 0$.

Proof. (a) Conditionally on $\Phi_\infty = \varphi$, the density of the observed path on $[q_0, q]$ with respect to the law of B is $\exp\{\langle P_q, \varphi\rangle - \frac{q-q_0}{2}\|\varphi\|^2\}$ (Girsanov, or a direct computation on finitely many increments). Bayes' formula gives ν_{q,P_q}, and differentiating $\mathcal{K}_q$ in p gives the conditional moments. Positivity of Cov_q holds because $\nu_{q,p}$ has a positive density. (b) For $s<q$, $\mathbb{E}_s[P_q - P_s] = (q-s)\,\psi_s$ since B has independent increments, and $\mathbb{E}_s\int_s^q \psi_u \mathrm{d}u = (q-s)\,\psi_s$ since ψ is a martingale. So W is a continuous martingale with $\mathrm{d}\langle W, W\rangle_q = I\,\mathrm{d}q$, and Lévy's theorem applies. (c) $\partial_q \mathcal{K} = -\frac{1}{2}\,\mathbb{E}_{\nu_{q,p}}\|\varphi\|^2$ and $\mathrm{Tr} D^2\mathcal{K} = \mathbb{E}_{\nu_{q,p}}\|\varphi\|^2 - \|D\mathcal{K}\|^2$. Differentiating (c) in p and applying Itô's formula along $\mathrm{d}P = D\mathcal{K}(P)\mathrm{d}q + \mathrm{d}W$, the drift of $D\mathcal{K}_q(P_q)$ cancels, which gives $\mathrm{d}\psi = \mathrm{Cov}\,\mathrm{d}W$. □

The observation P_q is not yet the field of Lecture 3. The posterior (11.6) is ν_N tilted by a Gaussian factor of precision $q-q_0$: the remaining randomness has precision $K_{q_0} + (q-q_0) = K_q$. This suggests the following.

Proposition 11.4. (The Polchinski process) *Let $\Phi_q := C_q P_q$ and $F_q := K_q(\psi_q - \Phi_q)$. Then*

$$\mathrm{d}\Phi_q = C_q^2 F_q\,\mathrm{d}q + C_q\,\mathrm{d}W_q, \qquad F_q = \mathbb{E}_q[-D V_N(\Phi_\infty)], \qquad \psi_q = \Phi_q + C_q F_q. \tag{11.9}$$

This is the FBSDE (5.7) of Lecture 3 for the profile (11.2): Φ is the interacting field at resolution q, F the effective force, ψ the forecast of Proposition 6.1, and $\mathrm{Cov}_q = H_q^{\mathrm{ex}}(\psi_q)$ (Proposition 11.5) is the response of Proposition 6.3. Under $\nu_N = \mu^C$ one has $F=0$, $\Phi_q = \psi_q$, and Φ_q is the Gaussian field X_q of Proposition 4.2, with law μ^{Q_q}.

Proof. $\mathrm{d}\Phi_q = -C_q^2 P_q\,\mathrm{d}q + C_q(\psi_q \mathrm{d}q + \mathrm{d}W_q) = C_q(\psi_q - \Phi_q)\mathrm{d}q + C_q \mathrm{d}W_q$, which is the first identity. For the second, integrate by parts in the density of $\nu_{q,p}$ with respect to Lebesgue measure, $\exp\{\langle p, \varphi\rangle - \frac{1}{2}\langle\varphi, K_q\varphi\rangle - V_N(\varphi)\}$: there is no boundary term, and

$$p = K_q\, D\mathcal{K}_q(p) + \int D V_N\,\mathrm{d}\nu_{q,p}. \tag{11.10}$$

At $p = P_q = K_q\Phi_q$ this reads $K_q(\psi_q - \Phi_q) = \mathbb{E}_q[-D V_N(\Phi_\infty)]$. In the Gaussian case the posterior is $\mu^{C_q}(\cdot - C_q P_q)$, so $\psi = \Phi$ and $\Phi_\infty - \Phi_q$ is independent of $\mathcal{F}_q$ with covariance C_q. Hence Φ_q is a Gaussian martingale with covariance $C - C_q = Q_q$. In general both the law of $(\Phi_\infty, (\Phi_q)_q)$ here and the law of $(X_\infty, (X_q)_q)$ under the measure $\mathbb{P}$ of Proposition 5.2 are the Gaussian law tilted by the same function $\mathrm{e}^{-V_N}/\mathcal{Z}_N$ of the terminal field, so they coincide. Since $P_q = K_q\Phi_q$, the filtration $(\mathcal{F}_q)$ is also the one generated by Φ. □

So at a fixed lattice the FBSDE of Lecture 3 is a *theorem*, with a Brownian motion W which is explicitly the innovation of a filtering problem. All objects of Section 6 have a statistical meaning:

Lecture 3	filtering at resolution q
field Φ_q	$C_q P_q$: posterior mean under the Gaussian prior μ^C
forecast ψ_q	posterior mean $\mathbb{E}_q\Phi_\infty$
response $H_q^{\mathrm{ex}}(\psi_q) = \mathrm{Cov}_q$	posterior covariance
force $F_q = K_q(\psi_q - \Phi_q)$	posterior mean of the terminal force $-D V_N(\Phi_\infty)$
Brownian motion W	innovation $P_q - \int\psi$
unresolved covariance $S_q = C_q$	prior precision plus observation precision, inverted

The observations need no ultraviolet regularity: P_q is defined for every q, with terminal resolution ∞ at fixed N. The lattice is the only ultraviolet cutoff, and $\Phi_q \to \Phi_\infty$ as $q \to \infty$ (Lemma 17.1).

11.4 The Wetterich formulation

Let $\Psi_q := \mathcal{K}_q^*$ be the Legendre transform of the log-partition function and

$$\Gamma_q(y) := \Psi_q(y) - \tfrac{1}{2}\langle y, K_q y\rangle, \qquad H_q^{\mathrm{ex}}(y) := (K_q + D^2\Gamma_q(y))^{-1}. \tag{11.11}$$

Comparing with (6.11), and using $S_q^{-1}=K_q$, Γ_q is, up to a function of q alone, the Γ_t of Lecture 3: $\Gamma_q+\frac{1}{2}\langle y,K_{q_0}y\rangle$ is the effective average action with the regulator $R_q=q-q_0$ (Exercise 11.1). The coordinate identity $P_q=D\Psi_q(\psi_q)$ reads

$$K_q\Phi_q=K_q\psi_q+D\Gamma_q(\psi_q),\qquad \text{i.e.}\qquad \Phi_q=\psi_q+C_qD\Gamma_q(\psi_q),\tag{11.12}$$

which is (6.5) with $F_q=D\Gamma_q$.

Proposition 11.5. (Wetterich equation with a mass regulator) *At fixed N, Γ_q is smooth and strictly convex after adding $\frac{1}{2}\langle y,K_qy\rangle$, and*

$$\partial_q\Gamma_q(y)=\tfrac{1}{2}\mathrm{Tr}H_q^{\mathrm{ex}}(y),\qquad \mathrm{Cov}_q{=}H_q^{\mathrm{ex}}(\psi_q).\tag{11.13}$$

Proof. $\mathscr{K}_q$ is smooth, strictly convex (its Hessian is a covariance) and superlinear (the posterior has full support), so Ψ_q is smooth and $D^2\Psi_q(D\mathscr{K}(p))=(D^2\mathscr{K}(p))^{-1}$. By the envelope theorem at fixed y and Proposition 11.3(c), $\partial_q\Psi_q(y)=-\partial_q\mathscr{K}_q(p)=\frac{1}{2}\mathrm{Tr}D^2\mathscr{K}_q(p)+\frac{1}{2}\|y\|^2$, with $p=D\Psi_q(y)$. Subtract $\partial_q\frac{1}{2}\langle y,K_qy\rangle=\frac{1}{2}\|y\|^2$. □

This is (6.12) with $\partial_qR=1$, up to the constant $-\frac{1}{2}\mathrm{Tr}C_q$ (which is infinite in the continuum and irrelevant here). It is an identity about the unknown measure ν_N, not a construction: to use it we would need estimates on Γ_q, which is what we want to prove. Lecture 3 suggests the strategy: since the FBSDE in forecast coordinates is

$$\mathrm{d}\psi_q=\mathrm{Cov}_q\,\mathrm{d}W_q,\qquad \mathrm{Cov}_q{=}H_q^{\mathrm{ex}}(\psi_q),$$

an *approximate* effective action gives approximate forecast coordinates, and the error should be governed by how badly the approximate action fails to solve (11.13).

11.5 Approximate forecast and error

Let $\bar\Gamma_q$ be an even functional on H_N, smooth in q and y, such that $\bar\Psi_q(y)=\frac{1}{2}\langle y,K_qy\rangle+\bar\Gamma_q(y)$ is uniformly convex, $D^2\bar\Psi_q\geq\underline{c}_qI$ with $\underline{c}_q>0$. Let $\bar{\mathscr{K}}_q=\bar\Psi_q^*$ and define the *approximate response*, the *defect* and its drift

$$\begin{gathered}\langle\text{gathered}|H_q(y):=(K_q+D^2\bar\Gamma_q(y))^{-1},\\ e_q(y):=\partial_q\bar\Gamma_q(y)-\tfrac{1}{2}\mathrm{Tr}[H_q(y)-H_q(0)],\qquad b_q(y):=-H_q(y)\,De_q(y).\rangle\end{gathered}\tag{11.14}$$

The subtraction of $H_q(0)$ only removes a function of q. Define the *approximate forecast* and the *error*

$$Y_q:=D\bar{\mathscr{K}}_q(P_q),\qquad Z_q:=\psi_q-Y_q,\qquad \Xi_q:=\mathrm{Cov}_q{-}H_q(Y_q).\tag{11.15}$$

Equivalently Y_q solves the approximate coordinate equation $K_q\Phi_q=K_qY_q+D\bar\Gamma_q(Y_q)$, i.e. (11.12) with $\bar\Gamma$ in place of Γ. If $\bar\Gamma=\Gamma$ then $Y=\psi$, $Z=0$, and e depends only on q, so $b=0$.

Theorem 11.6. (The FBSDE in approximate forecast coordinates) *At every fixed N,*

$$\begin{aligned}\mathrm{d}Y_q&=[b_q(Y_q)+H_q(Y_q)\,Z_q]\mathrm{d}q+H_q(Y_q)\,\mathrm{d}W_q,\\ \mathrm{d}Z_q&=-[b_q(Y_q)+H_q(Y_q)\,Z_q]\mathrm{d}q+\Xi_q\,\mathrm{d}W_q,\end{aligned}\tag{11.16}$$

with $Y_{q_0}=Z_{q_0}=0$, and $Y+Z=\psi$ is the martingale $\int\mathrm{Cov}\,\mathrm{d}W$.

Proof. By the Legendre identities $\partial_q \bar{\mathcal{K}}_q(p) = -\partial_q \bar{\Psi}_q(y) = -\frac{1}{2}\|y\|^2 - \partial_q \bar{\Gamma}_q(y)$ and $D^2 \bar{\mathcal{K}}_q(p) = H_q(y)$ at $y = D\bar{\mathcal{K}}_q(p)$. Hence

$$\partial_q \bar{\mathcal{K}}_q + \tfrac{1}{2}\mathrm{Tr} D^2 \bar{\mathcal{K}}_q + \tfrac{1}{2}\|D\bar{\mathcal{K}}_q\|^2 = -\big(\partial_q \bar{\Gamma}_q - \tfrac{1}{2}\mathrm{Tr} H_q\big) \circ D\bar{\mathcal{K}}_q = -e_q \circ D\bar{\mathcal{K}}_q + \tfrac{1}{2}\mathrm{Tr} H_q(0).$$

Differentiate in p: $\partial_q D\bar{\mathcal{K}} + \frac{1}{2}\mathrm{Tr} D^2(D\bar{\mathcal{K}}) + D^2\bar{\mathcal{K}}\, D\bar{\mathcal{K}} = -H\, De = b$. Now apply Itô's formula to $Y_q = D\bar{\mathcal{K}}_q(P_q)$ along $\mathrm{d}P = \psi\,\mathrm{d}q + \mathrm{d}W$:

$$\mathrm{d}Y_q = \big[\partial_q D\bar{\mathcal{K}} + \tfrac{1}{2}\mathrm{Tr} D^2(D\bar{\mathcal{K}})\big]\mathrm{d}q + D^2\bar{\mathcal{K}}\,(\psi_q \mathrm{d}q + \mathrm{d}W_q) = [b_q - H_q Y_q + H_q \psi_q]\mathrm{d}q + H_q\,\mathrm{d}W_q,$$

which is the first equation, since $\psi - Y = Z$. The second is $\mathrm{d}Z = \mathrm{d}\psi - \mathrm{d}Y$ and (11.8). At $q = q_0$, $P_{q_0} = 0$ and evenness give $Y_{q_0} = 0 = \psi_{q_0}$. □

Reading (11.16). The equation for Y is forward and explicit, once Z is known: its coefficients b, H are deterministic functions built from $\bar{\Gamma}$. The equation for Z is backward. Its terminal condition is not imposed but *proved*: if $\bar{\Gamma}_q$ matches the bare action as $q \to \infty$, then $qZ_q \to 0$ at each fixed N (Lemma 17.1). Its only source is the drift b, i.e. the gradient of the defect, transported by the propagator $\exp \int H$. So:

UV stability means that the explicit action $\bar{\Gamma}$ computes the forecast up to an error Z which is small uniformly in N.

For this we need: (i) a defect e whose gradient is integrable against the backward propagator, which is a deterministic renormalisation problem (Section 12); (ii) global coercivity of $\bar{\Gamma}$, so that $H_q(y)$ is a good noise coefficient for all fields (Section 13); (iii) a priori estimates for the coupled system (Lecture 5). Figure 11.1 summarises the logic of the three lectures.

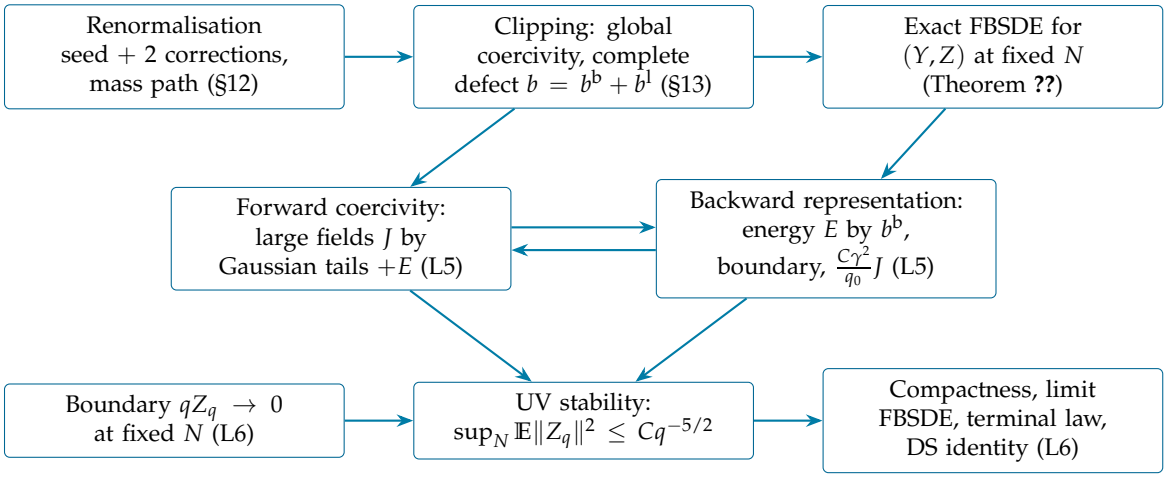


Figure 11.1. The structure of the construction. The two arrows between the forward and backward estimates form the closed loop of Lecture 5: neither estimate closes on its own.

Exercises

Exercise 11.1. (Same Γ) Show that Γ_q of (11.11) coincides, up to a constant, with the Γ_t of (6.11) for the profile (11.2), and that $\Gamma_q + \frac{1}{2}\langle y, K_{q_0} y\rangle$ is the effective average action Γ^{EAA} of Theorem 6.4. Check that (11.13) agrees with (6.12).

Exercise 11.2. (Gaussian case) Let $V_N(\varphi) = \frac{a}{2}\|\varphi\|^2$, $a > 0$. Compute $\mathcal{K}_q$, ψ_q, Cov_q and Γ_q explicitly, and check Proposition 11.4 and (11.13). Compare with Exercise 5.1.

Exercise 11.3. (Bayes for the whole path) Prove Proposition 11.3(a) for the whole observed path, not only for its value at q: show that P_q is a sufficient statistic.

Exercise 11.4. (One site) Take $N=1$ (one site, $\Delta=0$, $K_q=q$) and $\bar\Gamma_q(y)=\frac{\lambda}{4}y^4$. (a) Show that Y_q is the real root of $qY+\lambda Y^3=P_q$, and that $H_q(y)=(q+\gamma y^2)^{-1}$. Explain how large fields suppress the noise of Y. (b) Compute the defect $e_q(y)=-\frac12[(q+\gamma y^2)^{-1}-q^{-1}]$ and the drift $b_q(y)=-\gamma y\,(q+\gamma y^2)^{-3}$. Show that $|b_q(y)|\le\gamma q^{-3}|y|$, and compare with Proposition 13.2: on one site $a_2(q)=q^{-2}$ instead of $q^{-1/2}$, and no renormalisation is needed.

12 Renormalisation: why two of them

We now construct $\bar\Gamma_q$ on bounded fields. This section is deterministic: it uses no information on ν_N.

12.1 Scales and power counting

Two scales govern everything:

$$g_q:=\gamma q^{-1/2},\qquad r_q:=(q/\gamma)^{1/2},\qquad \gamma r_q^2=q. \tag{12.1}$$

g_q is the dimensionless coupling at resolution q: in $d=3$ the coupling λ has dimension of mass, and $q^{1/2}$ is the mass of the regulator. It is small at all scales if $g_{q_0}=\gamma q_0^{-1/2}$ is small; this is where the large mass enters. r_q is the field size at which the nonlinearity is comparable to the regulator: $D^2U(y)=\gamma y^2\approx q$ when $|y|\approx r_q$. Compare with the size of the free field at resolution q: $\mathrm{Var}X_q(x)=\int_{q_0}^q a_2\le Cq^{1/2}$, so typical fields have size $q^{1/4}=g_q^{1/2}r_q\ll r_q$.

We measure a k-tensor $T(x_1,\dots,x_k)$ (a kernel of a kth field derivative, a *vertex*) by its *rooted norm*

$$s(T)=\max_i\sup_{x_i}\int|T(x_1,\dots,x_k)|\prod_{j\neq i}\mathrm{d}x_j,$$

and a functional f by $\mathsf{S}_k(f)=\sup_{y\in\mathcal{B}_q}s\,(D^kf(y))$ on the *field ball*

$$\mathcal{B}_q:=\{y\in H_N:\|y\|_\infty<2r_q\}.$$

The natural size of the kth vertex of the loop corrections $\Gamma_q-U-\frac12\langle y,M_qy\rangle$ to the bare action is

$$\mathsf{S}_k\lesssim g_q\;q\;r_q^{2-k}. \tag{12.2}$$

For $k=2$ this says the Hessian of the corrections is g_q times the regulator q (while $D^2U\approx q$ at $|y|\approx r_q$); each extra field derivative costs r_q^{-1}. For $k=4$, $g_q\,q\,r_q^{-2}=\gamma g_q$: the quartic vertex of the corrections is γ times the small coupling. The whole construction is a bookkeeping of (12.2).

The basic loop estimate. Derivatives of $\frac12\mathrm{Tr}(K+D^2\Gamma)^{-1}$ are traces of closed chains $H\,D^{j_1+2}\Gamma\,H\cdots D^{j_p+2}\Gamma\,H$. A closed loop with p vertices, each carrying at least one external leg (vertices without external legs are resummed into H), and $p+1$ propagators H (two of them adjacent, H^2, once the trace is closed) is estimated by rooting one external slot, using two propagators in L^2 (each row has L^2 norm $\le(a_2)^{1/2}$) and the others in operator norm ($\le C/q$):

$$s(\mathrm{loop})\le C\;a_2(q)\;q^{-(p-1)}\prod_{i=1}^p s(T_i) \tag{12.3}$$

[FBSDE26]. The vertices of the bare action have $s\,(D^{j+2}U)\le Cqr_q^{-j}$ on the ball ($D^3U=2\gamma y$, $D^4U=2\gamma$). With $a_2\le Cq^{-1/2}$, a loop with k external legs built from them is $\le Cq^{1/2}r_q^{-k}$, and after the integration in q that produces Γ (which gains a factor q) it has size $q^{3/2}r_q^{-k}=g_q\,q\,r_q^{2-k}$: this is the origin of (12.2), the size of one loop. Replacing a bare vertex by a vertex of a correction of relative size ε multiplies the loop by ε (Exercise 12.1). *Every further loop gains one power of g_q.*

12.2 The seed: one loop

The first approximation is the one-loop formula of Exercise 6.7(b): replace $D^2\Gamma$ by $D^2U=\gamma y^2$ in the Wetterich equation and integrate from $q=\infty$. With the quadratic part subtracted this gives the *seed*

$$\begin{aligned}\mathcal{L}_q(y) &= \tfrac{1}{2}\mathrm{Tr}[\log\,(I+C_q^{1/2}\gamma y^2 C_q^{1/2}) - C_q^{1/2}\gamma y^2 C_q^{1/2}],\\ \partial_q\mathcal{L}_q(y) &= \tfrac{1}{2}\mathrm{Tr}[G_q(y)-C_q] + \tfrac{\gamma}{2}a_2(q)\|y\|_2^2,\end{aligned} \tag{12.4}$$

with $G_q(y)=(K_q+\gamma y^2)^{-1}$. $\mathcal{L}_q\le 0$, and its constant and quadratic Taylor coefficients at $y=0$ vanish. Its expansion in powers of the field,

$$\mathcal{L}_q(y)=\sum_{j\ge 2}\frac{(-1)^{j+1}}{2j}\mathrm{Tr}(C_q\gamma y^2)^j,$$

is the sum of one-loop diagrams with $2j$ external legs. The quartic term is the bubble $-\frac{\gamma^2}{4}\iint C_q(x,z)^2y(x)^2y(z)^2\,\mathrm{d}x\,\mathrm{d}z$, with $s(D^4\mathcal{L}_q(0))=6\gamma^2a_2(q)\le Cg_qqr_q^{-2}$, as predicted by (12.2).

12.3 Corrections and the mass

The seed solves the Wetterich equation only with D^2U inside the trace. Its defect is of relative size g_q compared with the one-loop flow $q^{1/2}r_q^{-k}$ (two loops). To improve it, let $\Pi f(y)=f(y)-f(0)-\frac{1}{2}D^2f(0)[y,y]$ remove the constant and quadratic Taylor coefficients (the *low jets*) of an even functional, let $M_q=\mu_qI+\Theta_q$ be a quadratic operator path (a running mass μ_q and an even, translation-invariant, centred kernel Θ_q, $\Theta_q1=0$), and set

$$T_q[M,h](y):=\tfrac{1}{2}\mathrm{Tr}[(K_q+M_q+D^2(U+h)(y))^{-1}-(K_q+M_q)^{-1}]. \tag{12.5}$$

The approximate action on the field ball is

$$\Gamma_q^{\mathrm{o}}(y)=U(y)+\tfrac{1}{2}\langle y,M_qy\rangle+h_q^{[2]}(y), \tag{12.6}$$

with the nonlinear part built by two *integrated* Wetterich steps,

$$h^{[0]}=\mathcal{L},\qquad h_q^{[j+1]}=\mathcal{L}_q-\int_q^\infty\Pi\,(T_s[M,h^{[j]}]-\partial_s\mathcal{L}_s)\mathrm{d}s,\qquad j=0,1, \tag{12.7}$$

so that $\partial_qh^{[j+1]}=\Pi\,T_q[M,h^{[j]}]$: each step inserts the previous approximation into the trace. The integrals run from ∞: the nonlinear part is fixed by the requirement that it vanishes at infinite resolution, where Γ_q must become the bare action (Theorem 6.4).

The quadratic part. The low jets are not iterated: they are handled exactly by the path M. The quadratic part of the flow at $y=0$ is, by evenness, $\frac{1}{2}D^2\mathrm{Tr}H(0)=-\frac{1}{2}\mathrm{Ctr}_{R_0^2}D^4\Gamma^{\mathrm{o}}(0)$, where $R_{0,q}=(K_q+M_q)^{-1}$ and Ctr_RT contracts two slots of T with the kernel R. Requiring that M solves this part of the flow exactly gives the *mass equation*

$$\partial_qM_q=B_q[M]:=-\gamma\mathrm{Tr}(R_{0,q}^2)\,I-\tfrac{1}{2}\mathrm{Ctr}_{R_{0,q}^2}D^4h_q^{[2]}(0), \tag{12.8}$$

(the source $B_q[M]$ is not the sampling Brownian motion B), with the *mixed boundary conditions*

$$\mu_{q_0}=0,\qquad \Theta_q\to 0\ \text{as}\ q\to\infty$$

(the latter in first spatial moment). The running mass is pinned at the physical scale q_0, while the nonlocal kernel is pinned at infinity. The *bare mass* is then

$$\mu_N = \lim_{q\to\infty} \mu_q = \int_{q_0}^{\infty} \beta_q \, \mathrm{d}q, \qquad B_q 1 = \beta_q 1, \tag{12.9}$$

and it is this number that enters the Gibbs measure (11.4). Note the order: the counterterm is an output of the deterministic flow, not an input.

Proposition 12.1. (The coefficient construction [FBSDE26]) *For q_0 large enough (depending on γ, not on N), the system (12.7)–(12.8) has a unique solution in a ball, with*

$$|\mu_q| \le C\gamma\sqrt{q}, \qquad v_1(\Theta_q) \le C\gamma^2 q^{-1/2}, \qquad \|\partial_q M_q\|_{\mathrm{row}} \le C g_q, \tag{12.10}$$

where $v_1(\Theta) = \int d(0,z)\,|\Theta(0,z)|\mathrm{d}z$. The seed obeys $\mathcal{N}_{k,q}(\mathcal{L}) \le C_k g_q q r_q^{2-k}$, $4 \le k \le 10$, and the corrections obey

$$\mathcal{N}_{k,q}(h^{[j]} - h^{[j-1]}) \le C_k \, g_q^{j+1} \, q \, r_q^{2-k}, \qquad 4 \le k \le 10 - 2j, \quad j = 1,2, \tag{12.11}$$

where $\mathcal{N}_{k,q}(f) = \mathsf{S}_k(f) + \sqrt{q}\; v_1(D^k f(0))$, and for a k-tensor

$$v_1(T) = \max_{i,a<b} \sup_{x_i} \int d(x_a, x_b)|T| \prod_{j\neq i} \mathrm{d}x_j.$$

The complete defect of Γ^{o} on the ball is $e_q^{\mathrm{o}} = \Pi\,(T_q[M, h^{[1]}] - T_q[M, h^{[2]}])$, and

$$\mathsf{S}_4(e_q^{\mathrm{o}}) \le C\gamma^5 q^{-3}. \tag{12.12}$$

At fixed N the bare mass exists, $|\mu_N| \lesssim \gamma \mathrm{Tr} C \lesssim \gamma N$, and $|\mu_N - \mu_q| \le C\gamma N^3/q$, $\|\Theta_q\|_{\mathrm{row}} \le CN\gamma^2 q^{-1/2} \to 0$ as $q \to \infty$.

Idea of the proof. (a) *Trace Lipschitz estimate.* If two corrections $h, \hat{h}$ obey the individual bounds (12.2) up to order $k+2$ and their difference obeys them with g_q replaced by ε_q, then $\mathcal{N}_k(T[M,h] - T[M,\hat{h}]) \le C\,\varepsilon_q\, q^{1/2} r_q^{-k}$. Expand D^k of the trace into loops; in each, telescope the difference onto one marked vertex or propagator and apply (12.3). (b) *Integration.* Each step gains one g; the integrand of (12.7) is $O(\gamma^{1+k/2} s^{-k/2})$ in the kth derivative, and that of the increment $h^{[j+1]} - h^{[j]}$ is $O(\gamma^{1+k/2} s^{-k/2} g_s^j)$, integrable at $s = \infty$ for $k \ge 4$. The low jets have been removed, so the lower derivatives are controlled by Taylor's formula from the fourth. (c) *Mass path.* Given M in the ball, the right side of (12.8) is computed from the corrections, and the map $M \mapsto (\mu, \Theta)$ is a contraction for large q_0: differences gain a factor g_{q_0} (for μ after integrating from q_0, using $\sup_x \log x/\sqrt{x} < \infty$). (d) *Defect.* The mass equation supplies exactly the quadratic jet of the flow, and $\partial_q h^{[2]} = \Pi T[M, h^{[1]}]$, so the defect is the trace difference between the last two approximations, of order g^3: by (a) with $\varepsilon_q = g_q^3$, $\mathsf{S}_4(e^{\mathrm{o}}) \le C g_q^3 q^{1/2} r_q^{-4} = C\gamma^5 q^{-3}$. □

Two technical points are hidden in this sketch. First, the trace estimate (a) needs more derivatives than it controls: the kth derivative of the trace involves vertices of order $k+2$. Ten derivatives of the seed leave eight after one step and six after two; this is why the number of steps must be finite and fixed in advance (an infinite iteration loses all derivatives). Second, the quadratic kernel Θ_q must be measured in a first spatial moment v_1, not in a second one: the estimate $\|\delta_z^2 C_q\|_{\mathrm{row}} \lesssim d(0,z)^2$, which a second moment would require, fails by a logarithm, and one uses $\|\delta_z^2 C_q\|_{\mathrm{row}} \lesssim q^{-1/2} d(0,z)$ instead [FBSDE26]. This is also why $\mathcal{N}_k$ contains a spatial moment at $y = 0$.

12.4 Two mass renormalisations

The bare mass μ_N diverges as $N\to\infty$. Its two divergent parts are the two divergent self-energy diagrams of Section 9.2.

- *The tadpole.* The first term of (12.8) integrates to $-\gamma\int_{q_0}^{\infty}\mathrm{Tr}R_{0,q}^2\,\mathrm{d}q\approx-\gamma\mathrm{Tr}C=-3\lambda\mathrm{Tr}C\sim-N$. This is Wick ordering: $\frac{\lambda}{4}\varphi^4-\frac{3}{2}\lambda c\varphi^2$ agrees, up to a constant, with $\frac{\lambda}{4}H_4(\varphi;c)$, with $c=C(x,x)$ ((9.5)).
- *The sunset.* The second term contracts the bubble contained in $D^4h^{[2]}(0)$ (the quartic coefficient of the seed) with R_0^2, and integrating in q produces the two-loop self-energy, which diverges like $\lambda^2\int_{|z|>1/N}|z|^{-3}\mathrm{d}z\sim\lambda^2\log N$. This is the sunset counterterm $\gamma_T\sim\log T$ of (9.7): with $\lambda_{\mathrm{L}3}=\lambda/4$ and $\gamma_T\int H_2=\gamma_T\int\varphi^2+\mathrm{const}$, the sunset part of μ_N is $2\gamma_T=6\lambda^2\int C^3\approx\frac{3\lambda^2}{8\pi^2}\log N$, positive. (The double-tadpole logarithms in the two terms of (12.8), the bubble with a tadpole insertion and the mass insertion in $\mathrm{Tr}R_{0,q}^2$, cancel each other.)

Power counting (Exercise 9.3) says that no other counterterm is needed. In the flow this is the statement that the integrals (12.7) of all vertices with $k\geq4$ legs converge at $q=\infty$: only the quadratic jet must be fixed by a boundary condition.

There is a subtlety: the running mass μ_q at fixed q has a limit as $N\to\infty$, but $\mu_N=\mu_\infty$ does not. One never interchanges these two limits. Also, the *whole* running kernel Θ_q is kept: replacing the sunset by its local part would leave a nonlocal remainder of the wrong size.

12.5 Two correction steps

Why two corrections, and not one or three? The answer is in the backward equation of (11.16). Freeze the noise and the large fields, and estimate the size of Z produced by a defect on the ball. The propagator of $\mathrm{d}Z=-HZ\,\mathrm{d}q+\dots$ from q to s is about s/q (since $H\approx C_q$ has norm q^{-1}), so

$$Z_q\approx\int_q^\infty\frac{s}{q}\,b_s\,\mathrm{d}s,\qquad|b_s|\lesssim\|H_s\|\,\mathsf{S}_4(e_s)\;r_s^3\lesssim s^{-1}\mathsf{S}_4(e_s)\;r_s^3,\tag{12.13}$$

where we used that the low jets of e vanish, so on the ball $|De|\leq\mathsf{S}_4(e)|y|^3/6$. After n correction steps the defect is the trace difference of the last two approximations, of relative size g^{n+1}: by the trace estimate,

$$\mathsf{S}_4(e_s)\lesssim g_s^{n+1}s^{1/2}r_s^{-4}=\gamma^{n+3}s^{-(n+4)/2},\qquad|b_s|\lesssim\gamma^{n+3/2}s^{-(n+3)/2},$$

and therefore

$$|Z_q|\lesssim\frac{\gamma^{n+3/2}}{q}\int_q^\infty s^{-(n+1)/2}\mathrm{d}s.$$

steps n	$\mathsf{S}_4(e_s)$	$\lvert b_s\rvert$	$\int_q^\infty\frac{s}{q}\lvert b_s\rvert\mathrm{d}s$
0 (seed only)	γ^3s^{-2}	$\gamma^{3/2}s^{-3/2}$	∞ (power)
1	$\gamma^4s^{-5/2}$	$\gamma^{5/2}s^{-2}$	∞ (logarithm)
2	γ^5s^{-3}	$\gamma^{7/2}s^{-5/2}$	$2\gamma^{7/2}q^{-3/2}$

With one step the bounded-field source produces a logarithmically divergent error; two steps are the minimum, and they suffice: $Z_q^{\mathrm{b}}=O(q^{-3/2})$, and the backward energy $\gamma^{-1}\int q^{1/2}|Z^{\mathrm{b}}|^2\mathrm{d}q$ is finite (Lecture 5). The general (fractional) theory gives the depth criterion $(n+2)\,\varkappa>(3+\varkappa)/2$, with subcriticality $\varkappa=\frac{1}{2}$ for Φ_3^4: $(2+2)\cdot\frac{1}{2}=2>\frac{7}{4}$, while $n=1$ gives $\frac{3}{2}<\frac{7}{4}$. This is a sufficient condition, which loses $\varkappa/2$; for Φ_3^4 the heuristic above shows that $n>1$ suffices, with $n=1$ borderline.

The two senses of "two renormalisations" are related. The mass counterterm must contain the sunset because the quartic vertex of the corrections, already of the seed, contains the bubble, and contracting it produces the sunset; and the second step is needed because the defect of the first one, the three-loop remainder, is borderline in the backward equation, exactly as the sunset is borderline in the mass.

Exercises

Exercise 12.1. (**Counting loops**) (a) Use (12.3) to show that a loop with p vertices $D^{j_i+2}(U+h)$, $j_i \geq 1$, each with $s \leq C q r_q^{-j_i}$, and $k = \sum j_i$ external legs has rooted norm $\leq C a_2(q)\ q\, r_q^{-k} \leq C q^{1/2} r_q^{-k}$. (b) Show that $q \cdot q^{1/2} r_q^{-k} = g_q q\, r_q^{2-k}$. (c) If one vertex is replaced by a vertex of size $\varepsilon\ q\, r_q^{-j}$, show that the loop gains the factor ε. Deduce the sizes g^{j+1} of the increments in (12.11).

Exercise 12.2. (**The bubble**) Compute $D^4 \mathcal{L}_q(0)$. Show that its rooted norm is $6\gamma^2 a_2(q)$ and its first spatial moment is $O(\gamma^2 q^{-1})$.

Exercise 12.3. (**Small Hessians are not enough**) On one site take $q = \gamma = 1$, $M = 0$, fixed small $\alpha, \delta > 0$, $\hat{h}_n(y) = \alpha\, n^{-2}(\cos(n\,y) - 1 + n^2 y^2/2)$, $h_n = \hat{h}_n + \frac{\delta}{12} y^4$. Show that both have small Hessians on $|y| < 2$, that $h_n - \hat{h}_n$ has bounded derivatives, but $D^4(T[h_n] - T[\hat{h}_n])(0) = 12\,(2\,\delta + \alpha\,\delta\, n^2 + \delta^2)$ is unbounded in n. Conclude that the trace estimate needs the individual higher derivative bounds.

Exercise 12.4. (**Why two steps**) Verify the table above. With one step, show that the bounded-field part of Z is of size $q^{-1} \log(T/q)$ on $[q, T]$, for an upper scale cutoff T.

13 The flow at large fields

13.1 The large-field problem

All the estimates of Proposition 12.1 hold on the field ball $\|y\|_\infty < 2 r_q$, and only there. But the forecast Y_q is a random field on N^3 sites, and nothing prevents it from being larger than r_q somewhere. The free field has typical size $q^{1/4} \ll r_q$, but it exceeds r_q with small, nonzero probability, and nothing a priori controls the interacting forecast. We need an action $\bar{\Gamma}_q$ defined and *uniformly convex for all fields*, which agrees with Γ_q^{o} on the ball.

A scalar model. The difficulty is already visible in the one-site seed, $\mathcal{L}_q(y) = \frac{1}{2}[\log(1 + \gamma y^2/q) - \gamma y^2/q]$. Its power series $\sum_{j \geq 2} \frac{(-1)^{j+1}}{2j} (\gamma y^2/q)^j$ has radius of convergence exactly r_q: the singularities are at $y = \pm i r_q$. Truncating it at any order produces a polynomial whose top coefficient has sign $(-1)^{j+1}$: for even $j \geq 4$ the truncation is unbounded below and destroys the convexity of $U + \mathcal{L}$ at $|y| \gg r_q$. The exact function is harmless: its logarithmic part grows only logarithmically, and $|\mathcal{L}_q''(y)| \leq \frac{9}{8}\gamma/q \leq 2\gamma/q$ for all y. In field theory we do not have the exact function, only bounds on the ball, obtained through an expansion of the trace in vertices (the corrections are not polynomials, but their estimates are expansions in the field). This is the *large-field problem*: perturbative control holds for $|y| \lesssim r_q$, and outside one needs a non-perturbative replacement that preserves the structure the estimates use.

The structure to preserve is coercivity, and it is carried by the bare quartic term. In the scalar model the coordinate equation $q\,y + \lambda\, y^3 = p$ has noise coefficient

$$H_q(y) = \frac{1}{q + \gamma y^2} \leq \frac{1}{q} \wedge \frac{1}{\gamma y^2}:$$

large fields are *more* elliptic and less noisy. A good large-field extension keeps U untouched and freezes only the correction.

13.2 Clipping

Let $\tau:\mathbb{R}\to\mathbb{R}$ be smooth and odd, $\tau(u)=u$ on $[-1,1]$, constant outside $[-2,2]$, with $0\le\tau'\le1$, and set

$$\kappa_q(y)(x):=r_q\,\tau\left(y(x)/r_q\right),\qquad \tilde h_q:=h_q^{[2]}\circ\kappa_q,\qquad \bar\Gamma_q:=U+\tfrac12\langle y,M_q y\rangle+\tilde h_q. \tag{13.1}$$

The clip acts pointwise in space and maps every field into the ball. The quartic and quadratic parts are not clipped. By the chain rule, with factors $|\kappa_q^{(b)}|\le C r_q^{1-b}$, the clipped correction obeys the same bounds (12.2) *globally*:

$$s\left(D^j\tilde h_q\right)\le C g_q q r_q^{2-j}\,(j=2,3,4),\qquad s\left(D^2\partial_q\tilde h_q\right)\le C g_q, \tag{13.2}$$

for all $y\in H_N$; the scale derivative includes the motion of the clip, $\partial_q\kappa_q=O(r_q/q)$.

Lemma 13.1. (Global coercivity) *For q_0 large, for all $y\in H_N$ and $q\ge q_0$,*

$$\langle \textit{gathered}|c\,(K_q+\gamma y^2)\le H_q(y)^{-1}\le C\,(K_q+\gamma y^2),$$
$$\|H_q(y)\|_{p\to p}\le\frac{1+C\,(g_q+g_q^2)}{q}\quad(1\le p\le\infty),\qquad \operatorname{diag}H_q(y)^2\le C a_2(q).\rangle \tag{13.3}$$

Idea. $H^{-1}=K_q+\gamma y^2+M_q+D^2\tilde h_q$, and $M_q+D^2\tilde h_q$ is small relative to K_q: $\|D^2\tilde h_q\|\le C g_q q$, and M_q is small after multiplication by C_q (the centred kernel is controlled by its first moment, Exercise 13.1). Write $H=(I+G\,(M+D^2\tilde h))^{-1}G$ with $G=(K_q+\gamma y^2)^{-1}$ and use the Neumann series and Lemma 11.1. □

So $\bar\Gamma_q$ is a globally convex action, and Theorem 11.6 applies.

13.3 The complete defect

On the set $\{\|y\|_\infty\le r_q\}$ the clip is the identity and the defect is $e^{\mathrm o}$, of size (12.12). Outside it, the clipped action does not solve the Wetterich equation even approximately in relative terms. Split the drift into its bounded-field and large-field parts,

$$b_q^{\mathrm b}(y):=-H_q(y)\;D\kappa_q(y)\;De_q^{\mathrm o}(\kappa_q y),\qquad b_q^{\mathrm l}:=b_q-b_q^{\mathrm b},$$

and let $\chi_q(y)=1_{|y|>r_q}$ (pointwise) and $t_q(y)=y\,\chi_q(y)$, the field restricted to the large-field set.

Proposition 13.2. (Complete source bounds [FBSDE26]) *Uniformly in $y\in H_N$, N and $p\in[1,\infty]$,*

$$\begin{aligned}
&\|b_q^{\mathrm b}(y)\|_p\le C\gamma^{7/2}q^{-5/2},\\
&\|b_q^{\mathrm l}(y)\|_p\le C\gamma q^{-3/2}(\|t_q\|_p+r_q^{-1}\|y^2\chi_q\|_p),\\
&\|b_q^{\mathrm l}(y)\|_2^2\le C\gamma^4 q^{-5}\|t_q\|_6^6.
\end{aligned} \tag{13.4}$$

Moreover $\|b_q(y)\|_p\le C\gamma q^{-3/2}\|y\|_p$.

The first bound is (12.13) with $n=2$: on the ball $|De^{\mathrm o}|\le \mathsf S_4(e^{\mathrm o})\,(2r)^3/6$, and $q^{-1}\gamma^5 q^{-3}r_q^3=\gamma^{7/2}q^{-5/2}$. The large-field part is as large as the bound $\gamma q^{-3/2}|y|=(g_q/q)|y|$ on the whole drift (no smallness beyond the factor g_q already present in b), but it is *supported on the large-field set* (up to nonlocal tails controlled by operator bounds). The last bound follows from the second since on $\{|y|>r\}$, $|t|+r^{-1}y^2\le 2r^{-2}|t|^3$, and $(\gamma q^{-3/2}r^{-2})^2=\gamma^4q^{-5}$.

Where the large-field error comes from. Compute the gradient of the defect of the clipped action and compare it with $D\kappa\, De^{\mathrm{o}}(\kappa y)$. With $z=\kappa_q(y)$, $J=D\kappa_q(y)$ (multiplication by $\tau'(y/r)$), $h=h_q^{[2]}$, $M'=\partial_q M_q$, and the unclipped response $H^{\mathrm{o}}=(K+M+\gamma z^2+D^2h(z))^{-1}$, the difference has four groups (seven terms):

i. the mass term $M'y-JM'z$: the quadratic part is not clipped, but its flow was computed at z;

ii. the motion of the clip, $JD^2h(z)\,\partial_q\kappa+(D\,\partial_q\kappa)^*\,Dh(z)$;

iii. $\gamma(y-Jz)\mathrm{diag}H^2$ and $\gamma Jz\mathrm{diag}(H^2-(H^{\mathrm{o}})^2)$: the quartic vertex sees y, the correction sees z, and the two responses differ by $\gamma(y^2-z^2)$ and by the chain-rule terms of the clipped Hessian, all supported on $\{|y|>r_q\}$;

iv. the contraction of $D^3(h\circ\kappa)-JD^3h(z)$ (the chain rule produces terms with κ'', κ''', supported on large fields) and of D^3h with $H^2-(H^{\mathrm{o}})^2$.

Each group vanishes if $\|y\|_\infty\le r_q$, and each is bounded by g_q times $|t|$ or $r^{-1}y^2\chi$, in L^p, by the same loop estimates ([FBSDE26]; Exercise 13.2 for the one-site version). There is no cancellation to exploit: the clipping error is genuinely of the size $(g_q/q)|y|$ of the whole drift on the large-field set. What saves us is that this set is rarely visited, and its occupation is controlled by the coercivity of the actual dynamics. Lecture 5 proves exactly this, with the weighted *large-field action*

$$\int_{q_0}^{\infty} q^{-3/2}\,\gamma\|t_q(Y_q)\|_6^6\,\mathrm{d}q,$$

(the t-part of the large-field action of Lecture 5), whose weight is matched to the last bound in (13.4): $q^{7/2}\cdot\gamma^4q^{-5}=\gamma^3\cdot(\gamma q^{-3/2})$, so $\int q^{7/2}\|b_q^1\|_2^2\mathrm{d}q\le C\gamma^3\int q^{-3/2}\gamma\|t_q\|_6^6\mathrm{d}q$, where $q^{7/2}$ is the weight with which $\|b^1\|_2^2$ enters the backward representation (Lecture 5).

The coordinate correction. Finally, write the approximate coordinate equation as

$$K_q\Phi_q=K_q(Y_q+\varrho_q(Y_q))+\tfrac{\gamma}{3}Y_q^3,\qquad \varrho_q(y):=C_q(M_qy+D\tilde h_q(y)),\tag{13.5}$$

using $DU(y)=\lambda y^3=\frac{\gamma}{3}y^3$. The correction is a small contraction, $\|D\varrho_q(y)\|_{p\to p}\le\varepsilon_q:=C(g_q+g_q^2)$, for all y. So, up to a small Lipschitz perturbation, *the forecast is the solution of the cubic elliptic equation $K_q(Y+\varrho)+\frac{\gamma}{3}Y^3=K_q\Phi_q$ driven by the field at resolution q*. This is the equation whose coercivity Lecture 5 exploits.

13.4 Summary

At the end of this lecture we have, at every lattice cutoff:

- a counterterm μ_N, a mass path M_q and a globally convex action $\bar\Gamma_q=U+\frac12\langle y,M_qy\rangle+h_q^{[2]}\circ\kappa_q$, all deterministic and constructed without any information on ν_N;
- the exact FBSDE (11.16) for the approximate forecast Y and the error Z, with noise coefficients bounded as in Lemma 13.1;
- a source $b=b^{\mathrm{b}}+b^{\mathrm{l}}$, whose bounded-field part is integrable in the backward equation thanks to the two correction steps, and whose large-field part is controlled by sixth powers of Y on the large-field set.

Lecture 5 turns these deterministic bounds into estimates on (Y,Z) uniform in N.

Exercises

Exercise 13.1. (Receiving a centred kernel) Let Θ be a translation-invariant, even kernel with $\Theta 1=0$. With T_z the translation by z, $(T_zf)(x)=f(x+z)$, and using $C_q\Theta=\frac12\int\Theta(z)\,(T_z+T_{-z}-2)\,C_q\,\mathrm{d}z$, show that $\|C_q\Theta\|_{\mathrm{row}}\le Cq^{-1/2}v_1(\Theta)$ if $\|(T_z+T_{-z}-2)\,C_q\|_{\mathrm{row}}\le Cq^{-1/2}d(0,z)$. Why is the absolute row mass of Θ not the right quantity?

Exercise 13.2. (The scalar clipping error) On one site, let $h(y) = -\beta \frac{\gamma^2}{4q^2} y^4$ (a quartic correction, β a fixed number) and $\bar{\Gamma}_q = \frac{\lambda}{4} y^4 + h(\kappa_q(y))$. (a) Show that $K_q + \bar{\Gamma}_q'' \geq q/2$ for all y if $\beta\gamma \leq c\, q^2$. (b) Show that $b_q = b_q^{\mathrm{b}}$ for $|y| \leq r_q$. For $|y| > 2r_q$ show that κ_q is locally constant, $b_q^{\mathrm{b}}(y) = 0$, and $b_q(y) = -\gamma y\,(q + \gamma y^2)^{-3}$, the drift of the pure quartic. (c) Conclude that on the large-field set the error is the whole drift of the trace term: it is comparable, for $|y| \approx 2r_q$, to the one-site bound $\gamma q^{-3}|y|$ on the whole drift (the analogue of $(g_q/q)|y|$, with $a_2(q) = q^{-2}$ in place of $q^{-1/2}$), but it vanishes on the ball. Compare with (13.4).

Exercise 13.3. (Large fields are less noisy) On one site with $\bar{\Gamma} = \frac{\lambda}{4} y^4$, show that the quadratic variation of Y is $\int H_q(Y_q)^2 \mathrm{d}q$ and that $H_q(y)^2 \leq (\gamma y^2)^{-2}$. Simulate the observation process for $\nu = \mathrm{e}^{-\lambda\varphi^4/4}\mu^{1/q_0}$ and plot P_q, $\Phi_q = P_q/q$, Y_q and ψ_q.

References

[BCG26] I. Bailleul, I. Chevyrev, M. Gubinelli. Stochastic observables and Wilson–Itô diffusions. In preparation.

[BFS83] D. Brydges, J. Fröhlich, A. Sokal. A new proof of the existence and nontriviality of the continuum φ_2^4 and φ_3^4 quantum field theories. *Comm. Math. Phys.* 91 (1983), 141–186.

[BG20] N. Barashkov, M. Gubinelli. A variational method for Φ_3^4. *Duke Math. J.* 169 (2020), 3339–3415.

[FBSDE26] M. Gubinelli. The massive Φ_3^4 field through forward–backward equations. Course notes, October 2026.

[FO76] J. Feldman, K. Osterwalder. The Wightman axioms and the mass gap for weakly coupled $(\Phi^4)_3$ quantum field theories. *Ann. Physics* 97 (1976), 80–135.

[GH21] M. Gubinelli, M. Hofmanová. A PDE construction of the Euclidean Φ_3^4 quantum field theory. *Comm. Math. Phys.* 384 (2021), 1–75.

[GJ73] J. Glimm, A. Jaffe. Positivity of the φ_3^4 Hamiltonian. *Fortschr. Phys.* 21 (1973), 327–376.